KAI-HSIN HUNG

Santa Clara, CA | (209) 453-5750 | k_hung2@u.pacific.edu | LinkedIn | GitHub | Portfolio

EDUCATION

UNIVERSITY OF THE PACIFIC, Stockton, California May 2026

Master of Science in Data Science, GPA: 3.96 / 4.00

Relevant Coursework: Machine Learning, Natural Language Processing, Relational Databases

MING CHI UNIVERSITY OF TECHNOLOGY, New Taipei City, Taiwan Jun. 2023

Bachelor of Science in Materials Science and Engineering

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Swift

ML & AI: PyTorch, TensorFlow/Keras, YOLOv8, Scikit-Learn, spaCy, Google Gemini API

Data & Visualization: Pandas, NumPy, Plotly, Matplotlib, PostgreSQL, PostGIS

Tools & Infrastructure: Docker, Google Cloud (Cloud Run, Firebase), FastAPI, Flask, React, Git, GitHub Actions (CI/CD)

EXPERIENCE

BirthdayMessaging.io, London, UK (Remote) Mar. 2025 – May 2025

Software Engineer Intern

- Synchronized 1,000+ unstructured contact records into a production database with zero manual entry, building a Python ingestion pipeline with a custom .VCF parser that parsed and validated inconsistent contact data.
- Engineered a cron-based task scheduler to query time-sensitive events and automate 100% of daily email dispatch, eliminating manual intervention.
- Designed idempotent state-management logic to prevent duplicate concurrent updates, maintaining 100% data integrity across message deliveries.

Formosa Plastics Group, Taiwan Nov. 2021 – Sep. 2022

Manufacturing Engineer Intern

- Organized and cleaned daily plant operational reports submitted by multiple teams, restructuring inconsistent, manually-entered records into clean, analytically-ready datasets that supported routine reporting.

PUBLICATION

K. Hung, G. Phung, F. Lorenzino, "GoPark: An AI-Powered Parking Recommendation System," *Intelligent Systems Conference (IntelliSys) 2026*, Springer Lecture Notes in Networks and Systems. (Accepted)

SELECTED PROJECTS

AI-Driven Toxicity Prediction & Data Curation (Capstone) May 2026

Python, Scikit-Learn, Gemini API, Pandas, REST APIs, NLP

- Built the linear baseline (0.42 mean per-target R^2 across 31 endpoints) that calibrated a 5-person, 7-model toxicology benchmark and motivated the non-linear models reaching 0.58 R^2, engineering Morgan-fingerprint and descriptor features from SMILES, dose, and duration on 30,106 records.
- Harmonized the DrugMatrix toxicogenomics database into a modeling-ready schema feeding the team's 30,106-record training set, integrating differential expression, clinical chemistry, and histopathology with PubChem SMILES/CASRN enrichment.
- Curated 2 of 10 hepatotoxicity sources (ATSDR, CEBS) in a team standardization effort, reconciling CAS and InChIKey identifiers into unified, de-duplicated records for modeling.
- Replaced manual abstract review by automating drug-hepatotoxicity literature mining over PubMed with the Gemini LLM, producing structured, confidence-scored extractions.

GoPark: AI-Powered Parking Recommendation System *(Accepted, IntelliSys 2026)*Dec. 2025

Python, YOLOv8, PyTorch, FastAPI, PostgreSQL/PostGIS, spaCy, SwiftUI

- Trained and deployed a YOLOv8-Nano object detection model achieving 97.2% mAP@0.5 with 5.4 ms inference latency on edge hardware, enabling real-time parking occupancy monitoring.
- Developed a Night-Time Optimization strategy using HSV augmentation, increasing vehicle detection recall by 5.3x under low-light conditions.
- Built a rule-based NLP parser with spaCy, achieving 96% constraint extraction accuracy at <2 ms latency (~500x faster than GPT-4) for interpreting natural language parking queries.

JobFit AI: Intelligent Resume Analysis System Jul. 2025

Python, Flask, Gemini API, PyMuPDF, React, Docker, Google Cloud Run

- Boosted user retention 30% with an AI resume analyzer that maps candidate skills to live job postings and surfaces gaps, built on the Gemini API and Adzuna feeds with fuzzy skill-matching.
- Developed a PDF parsing pipeline with PyMuPDF to extract structured text from unstructured resumes, normalizing diverse formatting styles into a standardized skill taxonomy for LLM analysis.
- Architected a full-stack streaming solution using Server-Sent Events (SSE) on Cloud Run and Firebase, reducing perceived response latency by 40%.